



TEXAS

The University of Texas at Austin
College of Liberal Arts
Department of Psychology

Sleep efficiency during the retention period predicts episodic memory reconstruction across young and old adults

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Introduction

- Sleep deprivation studies in young adults suggest that habitual sleep is essential for episodic memory consolidation¹, yet both sleep quality and memory tend to decline with age with substantial individual differences.²
- Relatively little is known about the impact of habitual sleep on episodic memory consolidation and supporting neural activity in young or older adults.
- Research question 1:** Do individual differences in post-encoding, retention, sleep quality predict those in episodic memory consolidation (i.e., retention) in young and/or older adults?
- Research question 2:** Do individual differences in sleep quality during the post-encoding retention predict those in neural activity supporting subsequent retrieval success in young and/or older adults?

Method

Participants

- 30 Young (M age = 21.2; 20 F); 33 Old (M age = 68.5; 17F)
- 58% White, 18% Asian, 31% Hispanic, 7% Black, 3% Multiracial, 4% Not reported

		Young Adults		Old Adults		p
		Mean	Sd	Mean	Sd	
Age (in years)		21.22	2.97	68.50	4.76	
Education (in years)		13.59	2.46	17.63	3.16	
Digit Span Time (s)		27.75	4.87	28.24	4.97	0.69
Trail Making	Total time (s)	61.65	18.05	78.67	34.24	<0.05
	Total error	0.90	1.42	0.47	0.93	0.15
CVLT	Free recall	13.53	1.80	10.68	3.53	<0.05
	Recognition	15.47	0.76	14.68	1.39	<0.05
Sleep PCA components	Duration	0.15	0.82	-0.21	0.81	0.09
	Efficiency	0.001	0.43	-0.03	0.38	0.75

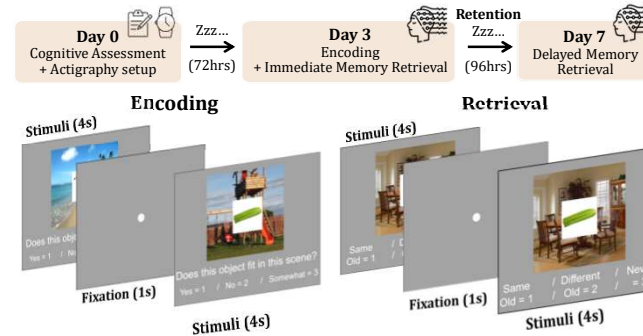
Sleep measures:

- Post-encoding (96hrs) was extracted from a 7-night sleep period collected using actigraphy.
- 2 PCA components: Sleep Duration and Sleep Efficiency

	Sleep Variable	Sleep Duration	Sleep Efficiency
Mean	Sleep Duration	0.87	0.01
	Sleep Onset Latency	0.14	0.90
	Sleep Efficiency	-0.16	-0.91
	Number of Awakening	0.84	0.39
	Wake After Sleep Onset (WASO)	0.89	0.19

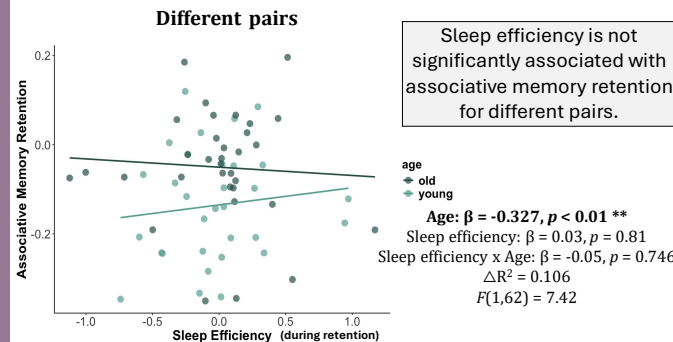
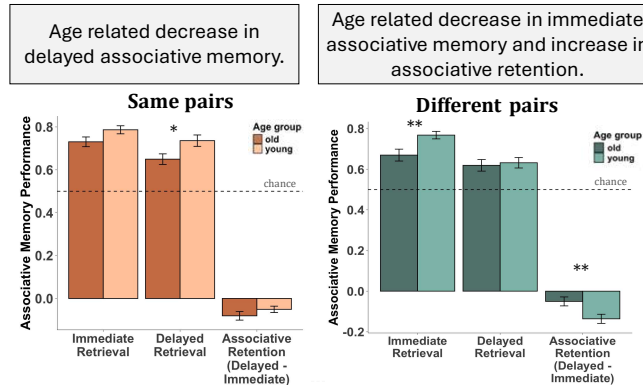
EEG measures: Recorded during encoding, immediate retrieval, and delayed retrieval memory tasks

Experimental Design

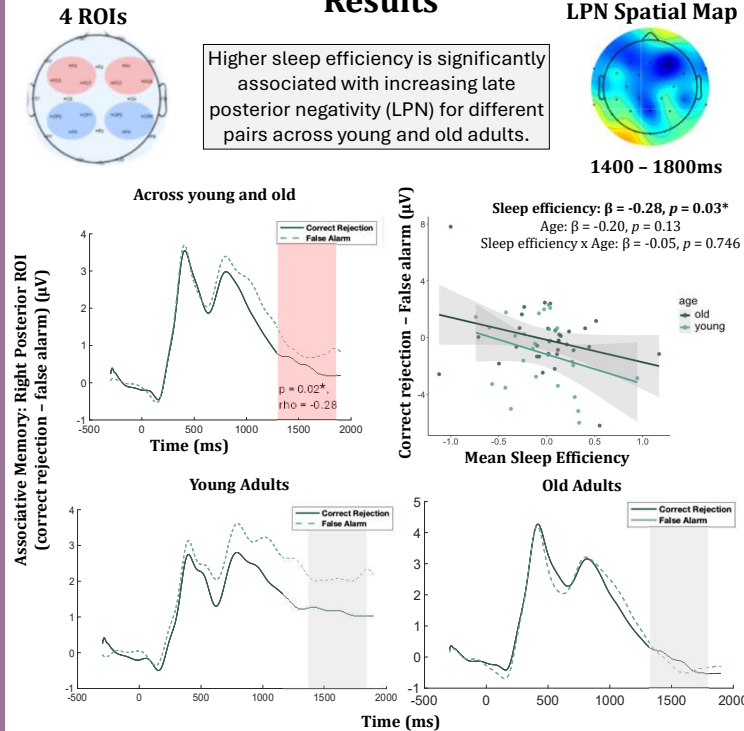


- Encoding:** Does the object in the middle fit the scene in the back?
- Retrieval:** Is this the old object with the same background = 1, the old object with a different background = 2, or a new object = 3?

Results



Results



Conclusion

- Our findings suggest that better sleep efficiency, particularly during the retention period, may enhance neural processes underlying episodic reconstruction and memory performance regardless of age.
- Given well known age related impairments in associative memory, these results suggest that quality sleep following encoding may be particularly important for preserving memory retention in older adults.

Acknowledgments

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References

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